

Shenglong Chen

Vancouver, BC | Open to Relocation

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UBC Cognitive Systems graduate specializing in **Distributed Systems, Heuristic Search, and High-Throughput Data Pipelines**. Proven track record in transforming unstructured regulatory data into production-ready ML features and architecting modular, scalable full-stack applications. Strong advocate for **Clean Code (SOLID)**, **Test-Driven Development (TDD)**, and **Cloud-native** architecture.

Technical Skills

Programming Languages: Python (Expert: Pandas, NumPy, Scikit-Learn, Flask, FastAPI), Java (Advanced: Spring Boot, JUnit, Swing), SQL (PostgreSQL, MySQL), R (Tidyverse), JavaScript, HTML/CSS

AI & Machine Learning: Large Language Models (LLMs), Prompt Engineering, Feature Engineering, Supervised/Unsupervised Learning, Heuristic Search Algorithms, Statistical Modeling, Multivariate Regression, Data Normalization, Neural Network Basics

Cloud & DevOps: Docker (Containerization), GitHub Actions (CI/CD), AWS (S3, EC2, Lambda), Linux/Unix Shell Scripting, Git/GitHub, Test-Driven Development (TDD), Unit Testing (PyTest, JUnit 5)

Software Engineering: Object-Oriented Design (OOD), SOLID Principles, Design Patterns (Strategy, Observer, Singleton), RESTful API Development, Microservices Architecture, Distributed Systems, Database Schema Design, Agile/Scrum

Data Engineering: ETL Pipeline Design, Multi-threaded Data Ingestion, Data Scraping/Mining, Parallel Processing (Multiprocessing), Data Visualization (Matplotlib, Seaborn, Tableau), Data Integrity Validation

Education

The University of British Columbia

Bachelor of Science, Major in Cognitive Systems

Vancouver, BC

Sep 2022 – Aug 2025

○ **Technical Electives:** Distributed Systems, Advanced Data Structures, Algorithmic Design, Artificial Intelligence, Statistical Modeling, Databases, Computer Systems, Software Construction, and Functional Programming.

Professional Experience

Simon Fraser University

Machine Learning Research Assistant

Burnaby, BC

May 2025 – Oct 2025

- Engineered scalable data extraction pipelines to structure 10k+ financial records; implemented **Multiprocessing** and **Concurrent Futures** in Python to achieve a **4x speedup** in data ingestion and eliminate I/O bottlenecks.
- Developed predictive models for institutional behavior using **Multivariate Regression**, identifying regulatory shifts with a **95% confidence interval** which provided the quantitative basis for a pending academic publication.
- Architected **Feature Engineering workflows** using Scikit-Learn to handle missing data imputation and categorical encoding, ensuring high model fidelity and reducing data leakage across high-dimensional training sets.
- **Optimized Database Performance:** Designed normalized PostgreSQL schemas and optimized indexing strategy, reducing query execution time by **35%** and enabling real-time analytical joins on million-row tables.
- **Collaborative Development:** Utilized **Git** for version control and spearheaded weekly **Sprint reviews** in an Agile environment, ensuring technical milestones were met 2 weeks ahead of the primary research deadline.

Key Technical Projects

Distributed RESTful Task Microservice: [Python / Flask / Docker / PostgreSQL]

01/2025 – 03/2025

- Architected a containerized backend using **Docker Compose** to isolate database and application layers, enabling 1-command deployment and 100% environment parity across local and cloud environments.
- Implemented **Secure Auth:** Integrated **JWT-based Authentication** and Bcrypt password hashing, successfully hardening the system against unauthorized access and credential-stuffing attacks during stress tests.
- Built **CI/CD Pipelines:** Leveraged **GitHub Actions** to automate unit testing and static code analysis, ensuring that 100% of merged code met production standards and passed all regression tests.
- **System Design:** Optimized read/write throughput via connection pooling and request validation logic, allowing the system to handle **200+** concurrent requests with sub-150ms tail latency.

AI Heuristic Solver Engine: [Java / AI Algorithms / OOP]

01/2024 – 04/2024

- Developed a high-performance **AI Agent** using **Backtracking** with **Heuristic Pruning** and **Memoization**, effectively navigating astronomical state-spaces by caching and reusing optimal sub-problem solutions.
- Refined **Algorithmic Complexity:** Achieved a **60% reduction in search time** by designing custom pruning functions based on puzzle-state entropy, allowing the solver to clear expert-level puzzles in under 2 seconds.
- Applied **Software Design Patterns:** Utilized the **Strategy Pattern** to swap search algorithms at runtime and the **Observer Pattern** for an event-driven GUI that synchronized visual state with background solver threads.
- **Memory Optimization:** Optimized the JVM heap usage by implementing custom card-state bitmasking, reducing memory footprint by **45%** during deep recursive searches.

Cloud-Ready Management Platform: [Java / SQL / JUnit / AWS]

09/2023 – 12/2023

- Engineered a full-stack **CRUD system** with a custom persistence layer; integrated **JUnit 5** to automate 50+ test cases, identifying and resolving 12 critical edge cases prior to final integration.
- **Robust Data Layer Architecture:** Designed a relational schema to handle complex multi-entity relationships, implementing **SQL triggers and constraints** to ensure 100% referential integrity and data atomicity.